Ayush Singh
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Projects

LibGen – Digital Library & Resource Management System | **GitHub** | **Live** **Oct 2025 – Nov 2025**

- Built secure **RESTful APIs** using **Node.js and Express**, implementing **JWT-based authentication** and **role-based authorization** via custom middleware.
- Designed optimized MongoDB schemas with indexes for users, resources, and downloads to improve query performance and scalability.
- Implemented real-time admin–user communication using Socket.IO, with persistent chat history and event-driven messaging architecture.
- Tech Stack: MERN STACK WEB DEVELOPMENT

MULTICORE SCHEDULER | **GitHub** | **Live** **Feb 2025 - Apr 2025**

- Designed and developed a web-based multicore CPU scheduling simulator to model task execution, priority-based scheduling, and core utilization.
- Implemented dynamic task preemption, automatic task reassignment on idle cores, and Gantt chart visualization for execution analysis.
- Improved scheduling efficiency by 30% using optimized data structures such as Binary Trees and Max Heaps.
- Tech Stack : HTML, CSS, JavaScript, Data Structures

Training

MERN STACK WEB DEVELOPMENT | **Cipher Schools** | **Certificate** **Jun 2025 – Jul 2025**

- Completed a month long summer training in MERN stack development, gaining hands-on experience in building full-stack web applications.
- Implemented core backend features including JWT-based authentication, password hashing using bcrypt.js, and database modeling with Mongoose.
- Developed an Online Notes Taking Application as the final evaluation, enabling secure user authentication and CRUD operations on notes.

Certifications

Cloud Computing **Oct 2025**
NPTEL—**Certificate**

Fundamentals of Network Communications **Sep 2024**
Coursera— **Certificate**

Introduction to Hardware and Operating Systems **Sep 2024**
Coursera—**Certificate**

Skills

Languages: Java, C++, JavaScript, Python
Technologies/Frameworks: SpringBoot, ReactJS, Node JS, Express JS, MongoDB, Tailwind
Domain Skills: Data Structures and Algorithms, Cloud Computing, DevOps, Agile

Achievements

- Achieved NPTEL Topper 5% Distinction in Cloud Computing.
- Solved 100+ problems in Leetcode across all categories.
- Consistently maintained over 9 CGPA throughout the semesters.

Education

Lovely Professional University **2023 – Present**
Bachelor of Technology- Computer Science and Engineering—CGPA: 9.09 *Phagwara, Punjab*

St Xavier's School **2021 – 2022**
12th—CGPA: 7.25 *Ballia, Uttar Pradesh*

Casterbridge School **2019 – 2020**
10th—Percentage: 94% *Ballia, Uttar Pradesh*